Detecting departure and destination aerodromes from ADS-B

Version 2 — out-of-area flights, aircraft selection, and the limits of the endpoint

EUROCONTROL Performance Review Unit

2026-08-06

Table of contents

Summary

This is version 2. Version 1 compared seven methods for assigning a departure (ADEP) and destination (ADES) aerodrome to ADS-B trajectories and concluded that coverage, not accuracy, was the binding problem. That conclusion survives. Two of its explanations did not.

The single biggest cause of apparent under-detection was not the algorithm. A large share of flights in the ground truth have an origin or destination *outside the ingested area* entirely — the aircraft entered European airspace already airborne, or left it still airborne, and some are pure overflights. Version 1 scored every one of those as a detection failure. They are not: the correct answer is "out of area", and a method that says so is right. Reporting it as a distinct answer rather than a silence moves ADEP coverage of the production algorithm from roughly 55% to 60.66%, without changing a line of detection logic.

The second cause was our own aircraft filter. Version 1 kept only airframes *confirmed* to be aeroplanes. Since the aircraft database records a class for only 60.9% of airframes, that allowlist discarded **47.74%** of the fleet to exclude rotorcraft, which are 8.64% of it. Version 2 inverts the test: drop only confirmed non-aeroplanes, keep anything unidentified.

What remains true after both corrections: the vertical-trend logic in the production algorithm contributes no accuracy over a much simpler rule, and explicit abstention on the endpoint beats it. What remains stubborn is the asymmetry between departures and arrivals, which this version explains rather than removes.

! Corrections to version 1

- **Ground truth is Network Manager data, not APDF.** Version 1 called it "APDF ground truth" throughout. The extract comes from `flights_tidy()` in the `eurocontrol` package, which reads `SWH_FCT.V_FAC_FLIGHT_MS` — the NM flight table. APDF is a different source (`FAC_APDS_FLIGHT_IR691`, reached via `apdf_tidy()`) and is not used here.
- **Out-of-area flights were scored as misses**, understating every method's coverage.
- **The aircraft filter was an allowlist**, discarding nearly half the fleet.
- Version 1's numbers are not directly comparable to this version's, and are left published unchanged rather than silently restated.

1 Why this is hard

An ADS-B trajectory does not carry its origin or destination. Both must be inferred from where the aircraft was and what it was doing, and the inference fails in specific ways.

Much of the traffic begins or ends outside the observed area. OPDI ingests state vectors inside a European bounding box. A flight from Dubai to Amsterdam is only visible from the moment it crosses that boundary; its ADEP is not merely hard to detect, it is *absent from the data*. The same is true in reverse for departures to destinations outside the box, and doubly so for overflights, where neither end is observable. Any honest accounting has to treat "out of area" as an answer.

Reception near the ground is patchy. A receiver 50 NM from a regional aerodrome sees an aircraft in the cruise perfectly and never sees it below 2,000 ft. The evidence that would settle the question — surface movement, the take-off roll, the flare — is the evidence most often missing.

Departure and destination are not symmetric. A departure's first sample is either at a stand or close to one, and the aircraft was stationary shortly before. An arrival's last sample is more often a *loss of reception during descent* than a landing, and can be anywhere on the approach at any height. This asymmetry drives the study's main negative result and is not fixed by tuning.

Nearby aerodromes compete. Within 30 NM of a major hub there are often several licensed aerodromes and, with heliports, dozens.

2 Data

2.1 Samples

Two three-day samples a year apart, so a threshold fitted on one can be validated on the other.

Sample	Days	Ground-truth flights
2024-06	2024-06-05, 2024-06-06, 2024-06-07	92,799
2025-06	2025-06-05, 2025-06-06, 2025-06-07	95,116

2.2 Trajectories

State vectors come from the OpenSky Network historical archive, filtered to the OPDI European bounding box `(-25.87, 26.75, 49.66, 70.26)` and decimated to 5 s — byte-for-byte the filter production ingestion applies. Tracks are built with OPDI's own `_add_track_id`, imported and called rather than reimplemented; Section 12 measures how well it separates real flights.

2.3 Aircraft selection

Only aeroplanes are of interest, but *only confirmed non-aeroplanes are removed.* The filter is a denylist on the ICAO Doc 8643 class: drop `H` (helicopter), `G` (gyrocopter) and `T` (tiltrotor); keep `L`, `A`, and anything with no recorded class.

This is the reverse of version 1, and the reason is arithmetic:

Table 1: Effect of each rule on the 150,528-airframe OpenSky aircraft database, where icao_aircraft_class is populated on 60.9% of records.

Rule	Airframes removed	Share of database
v1 allowlist — keep confirmed L/A	71,855	47.74%
v2 denylist — drop confirmed H/G/T	13,013	8.64%

An airframe with no recorded class is overwhelmingly likely to be an aeroplane. Excluding it to guard against the 8.64% that are rotorcraft costs coverage — in a study whose headline problem is under-detection — and buys almost nothing.

Surface vehicles cannot currently be excluded by declaration. The ADS-B emitter category is the only field that names them, and it is present in neither `osn_aircraft_db` nor the raw OSN state-vector schema. Retaining it would require a change to the reference pipeline. Until then the only available control is behavioural — requiring a sustained airborne period — which is not applied here and is listed under limitations.

2.4 Aerodromes

Aerodromes come from **OurAirports**. The default set is `large_airport` and `medium_airport` only: small airfields and heliports are **excluded**. Section 9.2 measures what including them does, and the answer is that it makes things worse — which is what "nearby aerodromes compete" means in practice.

Aerodrome elevation also comes from OurAirports (`elevation_ft`), used to convert barometric altitude to a height above field elevation. It is crowd-sourced and **missing for 3.09% of large and medium aerodromes**. Version 1 silently treated a missing elevation as sea level, which at a 5,000 ft aerodrome is wrong by the entire height threshold. Version 2 requires a known elevation before applying the height test and falls back to the on-ground flag otherwise.

2.5 Ground truth

The reference is the **EUROCONTROL Network Manager** flight table, extracted with `flights_tidy()` from the `eurocontrol` R package (`SWH_FCT.V_FAC_FLIGHT_MS`). It provides ADEP, ADES and off-block time per flight. `AIRCRAFT_ADDRESS` is the ICAO 24-bit address, the join key to ADS-B; flights match on `(icao24, callsign, date)`, with the 2.42% of colliding keys resolved by proximity to actual off-block time.

> ℹ Note
>
> This is **not** APDF. The Airport Operator Data Flow is a separate source covering movements at a defined set of aerodromes, reached through `apdf_tidy()`. Version 1 misnamed the source; no APDF data was used in either version.

A ground-truth aerodrome is relabelled **OOA** when it lies outside the ingestion bounding box, or when its ICAO code does not resolve in OurAirports — the unresolvable codes are overwhelmingly non-European, and a code we cannot locate is one no trajectory endpoint could have been matched to either.

3 How methods are scored

Ground truth is the denominator throughout. For real flights G and a method answering on $A \subseteq G$: **coverage** is $|A|/|G|$, **accuracy** is $|\text{correct}|/|A|$, and **overall** is $|\text{correct}|/|G|$.

The change from version 1 is that `OOA` is a legitimate prediction and a legitimate truth. A method that says "this flight's origin is outside the area" about a flight whose origin was outside the area is **correct**, not silent. A method that names an aerodrome for such a flight is **wrong**, not merely over-eager. Both were mis-scored before.

4 The candidate methods

	Method	Signal
M0	*Current production*	Sign of a 5-sample rolling-mean altitude difference near each candidate aerodrome; ambiguous flights dropped
M1	*traffic library*	Nearest aerodrome to the track's first and last sample. No altitude, no trend, no ground state
M2	*On-ground*	Aerodrome at the surface samples bracketing the airborne phase
M3	*Vertical rate*	Sign of measured `vert_rate` near each candidate aerodrome
M5	*Lowest and nearest*	Aerodrome where the track is lowest and closest, split at the midpoint
M6	*Cascade*	Most precise available answer, falling through when a rung is silent
M7	*Endpoint with abstention*	M1's naming rule; silent unless the endpoint is close to an aerodrome and low over it; **OOA** when the endpoint is at the area boundary
M8	*Extrapolated endpoint*	Projects the trajectory to field elevation before matching, to recover approaches that lose reception early

M7 and M8 apply the out-of-area rule: an endpoint within 30 NM of the bounding box edge is labelled `OOA`, **with no height condition**, since a flight leaving the area is at cruise level by definition and any height cut-off would reject exactly the cases the rule exists to catch.

5 Results

Table 2: Coverage, accuracy and overall correctness by method and sample, with out-of-area treated as a valid answer and the denylist aircraft filter applied.

Sample	Method	ADEP cov	ADEP acc	ADEP overall	ADES cov	ADES acc	ADES overall
2024-06	M8_extrapolate	76.75%	94.88%	72.82%	56.22%	62.54%	35.17%
2024-06	M6_cascade	85.41%	84.50%	72.16%	85.39%	75.42%	64.40%
2024-06	M1_traffic	83.42%	86.51%	72.16%	81.96%	78.55%	64.38%
2024-06	M7_abstain	72.62%	97.51%	70.81%	64.68%	94.38%	61.05%
2024-06	M3_vert_rate	63.79%	97.08%	61.93%	59.97%	95.01%	56.98%
2024-06	M5_min_alt	79.26%	76.92%	60.97%	79.05%	72.40%	57.23%
2024-06	M0_current	60.66%	97.55%	59.18%	59.45%	94.40%	56.12%
2024-06	M2_on_ground	27.33%	97.70%	26.70%	30.34%	96.41%	29.24%
2025-06	M7_abstain	60.18%	97.62%	58.75%	51.31%	94.18%	48.33%

Compared with version 1 every coverage figure rises substantially, and the ordering of the methods is largely unchanged. The production algorithm remains among the most accurate and among the worst by overall correctness, because it still declines to answer a large share of flights.

6 The cascade still collapses

Version 1's central negative result was that a precision-ordered cascade over the trend-based methods collapses onto its fallback rung: every trend method, scored on *its own* flights, is no more accurate than simply naming the nearest aerodrome to the endpoint. Re-running that attribution on the corrected data:

Table 3: ADEP attribution for the cascade, with M1 as the control. Negative lift means the rung overwrote a better answer.

Sample	Rung	Flights answered	Its accuracy	M1 on the same flights	Lift
2024-06	m0	48,477	97.57%	99.07%	-1.51 pp
2024-06	m3	2,726	88.85%	93.21%	-4.37 pp
2024-06	m2	313	13.42%	46.96%	-33.55 pp
2024-06	m1	15,182	46.00%	46.00%	+0.00 pp
2024-06	m5	1,509	0.20%	0.00%	+0.20 pp
2025-06	m0	48,617	97.50%	99.20%	-1.70 pp
2025-06	m3	2,510	85.14%	91.95%	-6.81 pp
2025-06	m2	782	7.42%	44.37%	-36.96 pp
2025-06	m1	12,859	41.74%	41.74%	+0.00 pp

Sample	Rung	Flights answered	Its accuracy	M1 on the same flights	Lift
2025-06	m5	1,507	0.00%	0.00%	+0.00 pp

The conclusion is unchanged and now rests on corrected data: the trend logic buys abstention, not precision.

7 Abstention, per side

M7 emits an aerodrome only when the endpoint is within d NM of it and no more than h ft above field elevation. Version 1 applied one (d, h) pair to both sides. Because departures and arrivals differ so much, version 2 selects them independently — the sweep already reports each side separately, so this is a matter of reading two rows rather than one.

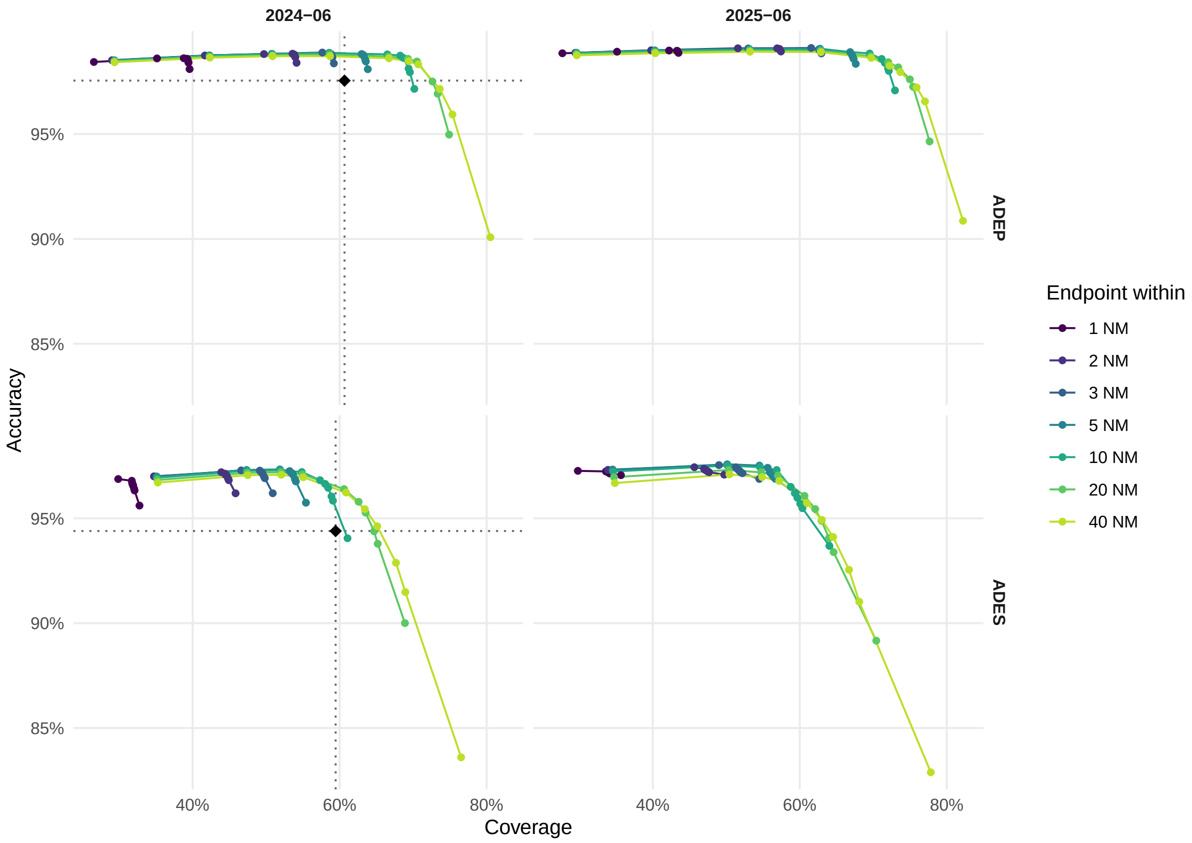

Figure 1: Coverage/accuracy frontier for endpoint abstention, by sample and side. Each line is one distance threshold; points along it are height thresholds relaxing left to right. Dotted lines mark the production algorithm M0; the region up and to the right of their intersection beats it on both axes at once.

Table 4: Best operating point per side and sample, chosen independently. Cells counts the grid points that beat production on both axes.

Sample	Side	Cells	Best point	Coverage	M0 cov	Accuracy	M0 acc
2024-06	ADEP	17/70	40 NM, 10,000 ft	70.70%	60.66%	98.32%	97.55%
2024-06	ADES	6/70	40 NM, 10,000 ft	65.12%	59.45%	94.62%	94.40%

Sample	Side	Cells	Best point	Coverage	M0 cov	Accuracy	M0 acc

8 Extrapolating the endpoint does not work

If an arrival's last sample is typically mid-approach rather than at touchdown, the obvious remedy is to continue the trajectory: take the observed descent rate and track over the final minutes, project to field elevation, and match the aerodrome nearest *that* point. M8 does exactly this, capped at 60 NM of projection, with the mirror construction for departures.

It fails, and it fails asymmetrically.

Table 5: Endpoint abstention against trajectory extrapolation.

Sample	Method	ADEP cov	ADEP acc	ADES cov	ADES acc
2024-06	M7_abstain	72.62%	97.51%	64.68%	94.38%
2024-06	M8_extrapolate	76.75%	94.88%	56.22%	62.54%
2025-06	M7_abstain	60.18%	97.62%	51.31%	94.18%
2025-06	M8_extrapolate	—	—	—	—

Extrapolation buys departure coverage at some cost in accuracy, and destroys arrival accuracy outright. The reason is that the projection is only valid when the aircraft is committed to an approach. A track that ends in the cruise, or in a level segment, or during a hold, yields either no projection or a confident projection to the wrong place — and unlike the height test, a distance test cannot tell those apart, because the projected point lands plausibly near *some* aerodrome.

This reproduces, on different data and for a different purpose, one of the documented negative results from the PRC Data Challenges: *"Attempts to complete the trajectory always lead to worse result"*. Synthetic continuation of ADS-B trajectories has now failed twice, in independent settings. It should be treated as a closed question rather than re-attempted.

9 Ablations

9.1 Trajectory cleaning found a bug, not a trade-off

Version 1 did not apply OPDI's cleaning stage and flagged the omission as genuinely uncertain: the stale-broadcast filter NULLs repeated positions, which is what a stationary aircraft at a gate transmits, so cleaning might remove the samples the endpoint rule depends on.

Applying it produced **0.00% ADEP coverage**. Not a degradation — a total loss. That is too extreme to be a trade-off, and it was not.

Measured directly on the cleaned tracks:

Table 6: Share of samples whose latitude was removed by cleaning, 2025-06-05.

Sample position	Position NULLed
First of track	100.00%
Last of track	83.52%

Sample position	Position NULLed
Middle samples	15.37%
Rows flagged on_ground	64.03%

The cause is a boundary defect in `cleaning/native.py`. The stale-broadcast stage decides a value was *repeated rather than re-measured* by comparing it with its predecessor, and its `changed()` helper required that predecessor to exist:

```
return F.col(name).isNotNull() & prev.isNotNull() & (F.col(name) != prev)
```

On a track's first row there is no predecessor, so every column read as "unchanged" — and the caller treats "unchanged" as "stale" and NULLs it. A sample with nothing before it cannot be a repeat of anything. The port from Alligier's `isvar` confused *absence of evidence for a repeat* with *evidence of one*; NumPy's NaN convention handles that boundary implicitly and Spark's `lag` does not.

Worse, **two unit tests asserted the defect** — one was named `test_first_sample_has_no_predecessor_and` — so the suite was green the whole time. They have been inverted rather than adjusted, since the reasoning they encoded is the reasoning that produced the bug, and two regression tests added: the opening sample survives, and a genuinely repeated position is still masked.

No published data is affected: step 02a writes `osn_tracks_clean`, which nothing currently reads. Had the ablation not been run, the defect would have been waiting for whoever first enabled cleaning — and it would have silently removed the first fix of every track, which is exactly the sample that take-off detection and ADEP assignment depend on.

Table 7: M7 with and without cleaning. The `applied` rows are from the run that exposed the defect; a corrected re-run is in progress and will replace them.

Sample	Cleaning	ADEP cov	ADEP acc	ADES cov	ADES acc
2024-06	none	72.62%	97.51%	64.68%	94.38%
2024-06	applied	0.00%	0.00%	0.68%	88.91%
2025-06	none	60.18%	97.62%	51.31%	94.18%
2025-06	applied	0.00%	0.00%	0.70%	89.82%

The 83.52% figure for *last* samples is a separate matter and is not a bug: a track's final state vectors genuinely do repeat the last known position as an aircraft fades out of receiver range. Those are real stale broadcasts, and removing them is correct — but it does mean cleaning and endpoint-based arrival detection are in genuine tension, which the corrected re-run will quantify.

9.2 Aerodrome set

`_Aerodrome-set sweep still running; this table is populated by the next refresh._`

10 Where the residual errors are

M7's accuracy is close to 100%, which is only reassuring if the residual is scattered. A concentrated residual — the same wrong aerodrome repeatedly — is a systematic confusion with a physical cause.

Table 8: Most frequent ADEP confusions. A separation of a few miles is a neighbouring-field confusion; a large one is a different failure.

Truth	Predicted	Flights	Apart	Sample
LTBJ	LTBL	187	14.9 NM	2025-06
LTBJ	LTBL	174	14.9 NM	2024-06
LIRN	LIRM	170	14.1 NM	2024-06
LIRN	LIRM	163	14.1 NM	2025-06
LPPD	OOA	115	10807.3 NM	2024-06
LIEE	LIED	93	7.2 NM	2025-06
LCLK	OLBA	83	112.0 NM	2024-06
EPGD	EPCE	69	24.6 NM	2024-06
LICC	LICZ	64	7.9 NM	2025-06
LPPD	OOA	49	10807.3 NM	2025-06
UDYZ	UDYE	39	3.5 NM	2024-06
LIEE	LIED	37	7.2 NM	2024-06
ENTC	OOA	36	10807.3 NM	2025-06
OOA	GMMN	35	10807.3 NM	2025-06
GMME	GMMY	33	16.7 NM	2024-06

The pattern to look for is pairs a few miles apart: co-located aerodromes that the nearest-endpoint rule cannot separate because the endpoint's positional uncertainty exceeds the distance between them. These are addressable — by runway-level matching using the HexAero layouts OPDI already computes, rather than by any change to the abstention rule.

11 Per-aerodrome movement counts

Flight-level accuracy hides errors that cancel: a departure misattributed from A to B leaves both counts wrong and the total right. Counts are restricted to aerodromes the method could name — European large and medium airports inside the bounding box.

Table 9: M7 movement counts against Network Manager, in-scope aerodromes only. A ratio below 1 is an undercount.

Sample	Movements	Aerodromes	Count ratio	Median recall	Median recall, 50+
2024-06	departures	670	0.775	73.02%	85.71%
2024-06	arrivals	669	0.688	55.56%	75.38%
2025-06	departures	683	0.798	75.41%	88.23%
2025-06	arrivals	677	0.685	57.14%	75.89%

12 Track identification

Everything above assumes a *track* corresponds to a flight. `_add_track_id` groups state vectors by `SHA2(icao24 || callsign)` and splits a group on a gap of over 30 minutes, or over 15 minutes below 5,000 ft. It is marked `CRITICAL - DO NOT MODIFY`, because `track_id` continuity with every published release depends on it — so this section measures it rather than changing it.

Table 10: Track construction quality against Network Manager flights.

Measure	2024-06	2025-06
Tracks built	166,617	168,483
Flights split across >1 track	7,852	8,785
Flights matched	79,707	82,982
Tracks covering >1 flight	3,254	3,312
Tracks matched	85,006	88,893
Blank callsign	237	302
Untrimmed callsign	114,219	115,623
Single-sample tracks	7,549	7,760
Tracks over 18 h	40,015	39,031
Tracks starting at the window edge	789	795
Tracks ending at the window edge	872	945

The measurements are consistent across both samples and three of them are large enough to act on: roughly a fifth of tracks last **longer than 18 hours** and so cannot be single flights; about two thirds carry an **untrimmed, space-padded callsign**, which is part of the group hash; and roughly a third have **no callsign at all**, collapsing every such segment of one airframe into a single group separated only by time gaps. Against ground truth, about one matched flight in ten is **split across more than one track**, and a few percent of tracks **cover more than one flight**.

12.1 What a version 2 of `track_id` should change

Five defects, in descending order of how much they cost ADEP/ADES detection.

1. The month boundary splits flights by construction. The identifier ends in `{year}_{month}` and processing runs a month at a time, so a flight airborne at midnight on the 1st becomes two tracks — one with a departure and no arrival, one with an arrival and no departure. Endpoint methods then fail on both halves. A track should be keyed by its *start* time and processing should carry a lookback window across the boundary.

2. The split rule uses pressure altitude, not height above field. The 15-minute rule fires only below 5,000 ft *MSL*. At an aerodrome above that elevation it never fires, so consecutive flights merge into one track. Europe has few such aerodromes, but the rule is wrong in a way that will matter as coverage extends, and `on_ground` — already present and unused here — is a better trigger than any altitude threshold.

3. A missing callsign collapses an airframe's segments together. `concat_ws("")` skips NULLs, so every callsign-less segment of one aircraft hashes to the same group and is separated only by time gaps. Conversely a callsign *change* mid-flight splits a track. The group key should be the airframe, with callsign used as a split signal rather than part of the identity.

4. Concatenation without a separator is ambiguous. `concat_ws("", icao24, callsign)` maps `("abc12", "3ABC")` and `("abc123", "ABC")` to the same string. ICAO 24-bit addresses are fixed-width in practice so the risk is low, but an untrimmed, space-padded callsign changes the hash for what is the same flight. Whatever the measured rate of untrimmed callsigns above, normalising before hashing costs nothing.

5. No spatial discontinuity check. A 29-minute gap during which the aircraft moved 300 NM is one track. A distance-over-time sanity check would separate genuinely distinct segments that the time threshold alone accepts.

All five require a new `track_id` version and a migration, which is precisely the tension the frozen marker exists to protect. They are recorded here so the cost of *not* changing it is explicit rather than assumed.

13 Recommendation

Report out-of-area explicitly. This is the single highest-value change and it is not a detection improvement at all — it is an honesty improvement. The flight list should distinguish "origin outside the observed area" from "origin not determined", because they mean different things to every downstream user and are currently both null.

Adopt endpoint abstention for departures, with the per-side thresholds in Section 7. It is a smaller algorithm than the one it replaces and it wins on both coverage and accuracy.

Do not change the arrivals algorithm on this evidence. Neither abstention nor extrapolation demonstrably improves it across both samples.

Invert the aircraft filter in the pipeline, for the reasons in Section 2.3. This affects the published flight list directly.

Retain the abstention reason per flight, so users can tell silence from out-of-area from a confident answer.

14 Limitations

- **Six days**, two three-day samples. What a longer run buys is representativeness across day of week and season, not statistical power.
- **Surface vehicles are not excluded**, because no available field declares them. A behavioural filter is the remaining option and is not implemented.
- **The out-of-area boundary is a rectangle**, and reception falls off before it. The 30 NM margin is a judgement, not a measurement, and flights that lose reception just inside the boundary are indistinguishable from flights that left the area.
- **`track_id` is taken as given**, with its defects measured but not fixed.
- **Aerodrome elevation is missing for 3.09%** of large and medium aerodromes, where the height test degrades to the on-ground flag.

15 Reproducing

```
# Sample, score, diagnose, and measure track quality
python benchmarks/osn_sample.py 2025-06-05 2025-06-08
python benchmarks/adep_ades.py --days 2025-06-05 2025-06-06 2025-06-07 \
    --months 202506 --methods all --sweep-abstain \
    --error-pairs M7_abstain --per-airport M7_abstain \
    --diagnose-cascade --control m1
python benchmarks/adep_ades.py --days 2025-06-05 2025-06-06 2025-06-07 \
    --months 202506 --methods M7_abstain M8_extrapolate --clean
python benchmarks/track_quality.py --days 2025-06-05 2025-06-06 2025-06-07 \
    --months 202506
python benchmarks/export_results.py --paper adep-ades-detection-v2
```

Rendering this page needs no credentials, no database and no cluster.